
Speech Emotion Recognition on the IEMOCAP dataset

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Abstract

This project presents a comparative study on speech emotion recognition using the Interactive Emotional Dyadic Motion Capture dataset, aiming to accurately identify emotion classes from speech signals. The study evaluates a machine learning model (Random Forest classifier) as a baseline alongside two advanced state-of-the-art models: a deep learning model (Bidirectional Long Short-Term Memory network with attention mechanism) and a transfer learning model (Wav2Vec2.0). The models are assessed on a set of hard-labelled emotion classes using metrics such as accuracy and unweighted average recall. The deep learning model demonstrates superior performance with an unweighted average recall of 58.00% on the 4-class task and 31.00% on the more challenging 8-class task. These findings emphasize that modelling temporal speech dynamics are key to improving accuracy with emotion recognition, and recurrent neural networks are particularly effective in capturing these sequential dependencies. Future advancements should focus on continuous emotion representation, enhanced pre-processing, and multimodal integration to better capture emotional nuances and support real-world applications.

1 Introduction

Speech Emotion Recognition (SER), a combination of understanding spoken content from speech signals, combined with capturing the underlying sentiment, has undergone a significant evolution, from being useful in niche applications to enabling a huge revolution in the field of speech analytics, leading to wider implications in health, security, robotics, user experience, and much more (Schuller 2018). One of the key ideas behind SER is that speech waveforms are different for each emotion displayed, regardless of the speaker, and using this unique waveform tied to each emotion can help us recognise the emotion behind a specific speech content (Abdumohsin et al. 2020).

1.1 Purpose and Goals

The purpose of this project is to apply learned concepts and implement Machine Learning (ML), Deep Learning (DL), and Transfer Learning (TL) techniques for recognizing human emotions from speech signals. By exploring how emotional cues are embedded in vocal patterns, the project aims to develop working pipelines capable of recognizing discrete emotion categories, referred to as hard-labelled emotions throughout this report, from speech data with significant accuracy and Unweighted Average Recall (UAR). UAR is particularly useful in this context, as it mitigates the impact of class imbalance by giving equal importance to all emotion categories during evaluation. This involves selecting and pre-processing a suitable data set, extracting relevant audio features, applying and evaluating different learning architectures, and finally conducting performance assessments using appropriate evaluation metrics. Among the models evaluated, the BiLSTM-based architecture achieved the best results. It reached an accuracy of 58.00% and a UAR of 58.00% on the 4-class task. On the more challenging 8-class task, it attained an accuracy of 42.19% and a UAR of 31.00%. These results demonstrate the effectiveness of deep temporal modelling in capturing the emotional content of speech.

1.2 Report Outline

The report continues with **Section 2 (Background)**, which reviews key concepts in SER, including audio features, emotional models, and relevant DL methods. **Section 3 (Methodology)** describes the dataset selection, pre-processing pipeline, model design choices, and evaluation metrics. **Section 4 (Implementation)** details the tools, frameworks, model architecture, training procedures, and preliminary outputs. **Section 5 (Related Work)** surveys existing literature on SER, highlighting established approaches and techniques. Finally, **Section 6 (Conclusions and Further Enhancements)** summarises the results, reflects on the challenges and contributions, and proposes future directions such as expanding to multimodal inputs or optimizing model generalization.

2 Background

SER is the task of recognizing emotions such as Happy, Sad, Angry, or Neutral from speech signals. Emotions affect various acoustic features of speech – like pitch, tone, and rhythm – which can be analyzed to infer the speaker’s emotional state (El Ayadi et al. 2011). SER has applications in mental health support, customer service, and emotionally aware virtual assistants (Schuller 2018).

Two main models are used in SER. The discrete emotion model treats emotions as distinct categories, while the dimensional model represents them along continuous scales like arousal (calm to excited), valence (unpleasant to pleasant), and dominance (Russell 1989). The dimensional model allows for capturing more subtle or mixed emotions, while the discrete model is easier to implement and is widely used in benchmark datasets such as RAVDESS (Livingstone & Russo 2018) and CREMA-D (Cao et al. 2014).

Traditionally, SER used ML models where handcrafted features like Mel-Frequency Cepstral Coefficients (MFCCs) were classified using algorithms like SVMs (Schuller & Rigoll 2009). However, these methods struggle to capture the temporal dynamics of speech. More recent models rely on DL, especially Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which are well-suited for sequential data and can learn patterns over time without manual feature engineering (Trigeorgis et al. 2016).

In this project, we conduct a comparative study of three different models for SER: a classical ML model using Random Forest (RF), a DL model based on a Bidirectional LSTM with attention (BiLSTM), and a TL model leveraging a pre-trained Wav2Vec2 architecture. This comparison aims to objectively evaluate the strengths and limitations of each paradigm in capturing emotional cues from speech, providing insights into their relative effectiveness under consistent experimental conditions.

A key motivation of this study is to observe the practical trade-offs in performance, interpretability, and computational requirements for each of the chosen model paradigms, thereby providing clearer guidance on the suitability of each model type for real-world applications where hard-labelled emotions are considered.

3 Methodology

3.1 Dataset Description

The dataset used in this study is the Interactive Emotional Dyadic Motion Capture dataset (IEMOCAP) (Busso et al. 2008), accessed via the HuggingFace Datasets hub (AbstractTTS/IEMOCAP) (AbstractTTS 2024). It consists of approximately 12 hours of audiovisual data, collected from ten actors in both scripted and improvised scenarios. Each utterance is annotated in three different ways: a hard-label of the emotion category in words (e.g., Happy, Sad), soft probabilistic values indicating the distribution over various emotions (e.g., Happy 0.97, Sad 0.02, Frustrated 0.01), and dimensional representations using valence, arousal, and dominance scores. Furthermore, the data set includes acoustic features such as pitch, energy, and speech rate, which are relevant for speech-based emotion recognition.

In this study, we focus exclusively on four primary target emotions (Yang et al. 2021) from the IEMOCAP dataset – Happy (including Excited), Sad, Angry, and Neutral – to simplify the task and enable more meaningful analysis. Each utterance in this selected subset is assigned a single hard-label corresponding to one of these target emotions.

3.2 Model Architectures

We implemented and compared three distinct model architectures to perform emotion recognition on utterances labelled with one of the target emotions. The first model is a classical ML model based on RF; the second is a DL model using a Bi-LSTM with attention (Yu & Kim 2020); and the third leverages TL through a fine-tuned Wav2Vec2-based architecture pre-trained on large-scale speech data.

RF Classifier: The RF classifier is an ensemble learning method that operates by constructing a multitude of decision trees during training and outputting the class that is the mode of the classes predicted by individual trees. It leverages the concept of bagging, where each tree is trained on a random subset of the data and features, thereby reducing variance and improving generalization. Due to its ability to handle high-dimensional data and resist overfitting, it can be effective for speech-related tasks where statistical acoustic features often exhibit variability and redundancy. Formally, the Random Forest algorithm can be described as follows:

Let the training dataset be $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^d$ is the feature vector and $y_i \in \{1, \dots, K\}$ is the corresponding class label.

A Random Forest constructs T decision trees $\{h_t(x)\}_{t=1}^T$, each trained on a bootstrap sample $\mathcal{D}_t \subset \mathcal{D}$ drawn with replacement. At each split in a tree, a random subset of $m < d$ features is selected to determine the best split, introducing feature-level randomness.

For a given test input x , each tree h_t predicts a class label $\hat{y}_t = h_t(x)$. The final prediction $\hat{y}$ of the forest is determined by majority voting:

$$\hat{y} = \arg \max_{k \in \{1, \dots, K\}} \sum_{t=1}^T \mathbb{I}[h_t(x) = k]$$

where $\mathbb{I}[\cdot]$ is the indicator function.

BiLSTM with Attention: The DL model is a BiLSTM, a bidirectional RNN that captures temporal patterns in both directions. It uses two stacked layers with 64 hidden units per direction to generate contextualised representations. Layer normalization is applied, followed by an attention mechanism that computes a weighted sum across time steps. The context vector then passes through dropout and a dense layer to produce class logits. This combination of temporal modelling and attention suits emotion recognition tasks, as illustrated in figure 1. The model is trained with cross-entropy loss.

Formally, let an input utterance be represented as a sequence of feature vectors $X = [x_1, x_2, \dots, x_T]$, where each $x_t \in \mathbb{R}^d$ corresponds to the feature vector at time step t , and T is the number of time steps.

The BiLSTM processes the sequence in both forward and backward directions:

$$\vec{h}_t = \text{LSTM}_{\text{fwd}}(x_t, \vec{h}_{t-1}), \quad \overleftarrow{h}_t = \text{LSTM}_{\text{bwd}}(x_t, \overleftarrow{h}_{t+1})$$

The hidden state at time t is formed by concatenation:

$$h_t = [\vec{h}_t; \overleftarrow{h}_t] \in \mathbb{R}^{2h}$$

An attention mechanism computes a context vector c as a weighted sum over time steps:

$$\alpha_t = \frac{\exp(w^\top \tanh(W h_t + b))}{\sum_{i=1}^T \exp(w^\top \tanh(W h_i + b))}, \quad c = \sum_{t=1}^T \alpha_t h_t$$

where W, w, b are learnable parameters. The context vector c is passed through dropout and a fully connected layer to produce output logits z :

$$z = W_o \cdot \text{Dropout}(c) + b_o, \quad \hat{y} = \text{Softmax}(z)$$

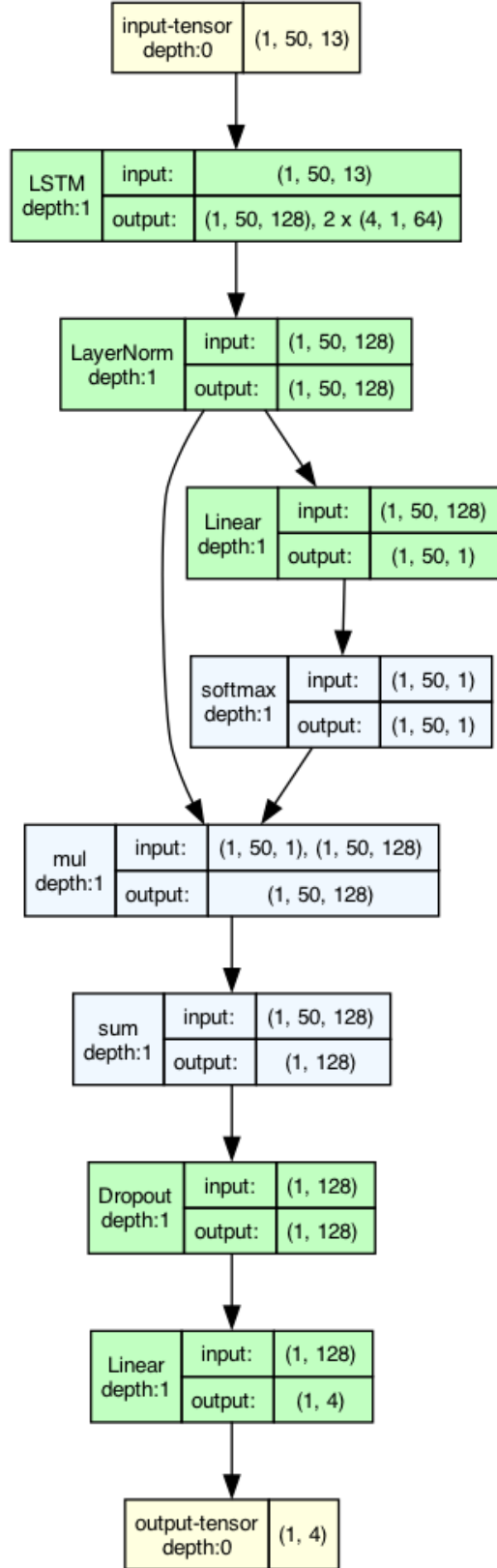


Figure 1: Architecture diagram of the BiLSTM-based Emotion Recognition Network. The model includes two bidirectional LSTM layers, followed by layer normalization, an attention mechanism to compute context vectors, dropout for regularization, and a final fully connected layer for classification.

Wav2Vec2 Feature Extractor + Feed-Forward Classifier: The TL model chosen is based on the pre-trained ‘facebook/wav2vec2-large’ model (Baevski et al. 2020)(Pepino et al. 2021), which is used as a frozen feature extractor to leverage its robust representations of raw speech. For each utterance, the hidden states from the final layer are mean-pooled to obtain a fixed-length 1024-dimensional feature vector. This is passed through a feed-forward classifier comprising two hidden layers with 512 and 256 units, each followed by batch normalization, ReLU activations, and dropout with a rate of 0.6. The final linear layer outputs class logits, which are passed through a log-softmax for training using the negative log-likelihood loss. While an alternative strategy would be to fine-tune the entire Wav2Vec2 model end-to-end, training such a model is significantly more computationally expensive and memory intensive. Freezing the encoder allows the model to benefit from pre-trained acoustic features while keeping training efficient and tractable for smaller datasets and limited hardware setups. For the mathematical representation:

Let $x \in \mathbb{R}^T$ denote the raw speech signal. The frozen Wav2Vec2 encoder $\mathcal{E}$ maps x to latent representations $\mathbf{H} \in \mathbb{R}^{L \times 1024}$. Mean pooling yields a fixed-length vector:

$$\mathbf{h} = \frac{1}{L} \sum_{i=1}^L \mathbf{H}_i$$

This is passed through a feed-forward classifier $\mathcal{F}$ with ReLU, batch normalization, and dropout layers, producing class logits $\mathbf{z} \in \mathbb{R}^C$:

$$\mathbf{z} = \mathcal{F}(\mathbf{h})$$

Training uses negative log-likelihood loss over log-softmax outputs:

$$\mathcal{L} = -\log \left(\frac{\exp(z_y)}{\sum_{c=1}^C \exp(z_c)} \right)$$

where y is the ground-truth label.

4 Evaluation

4.1 Data Pre-Processing and Feature Extraction

The IEMOCAP audio samples were loaded using the Hugging Face Datasets API, and only utterances labelled with one of the target emotions were retained. Each audio waveform was converted into a 13-dimensional MFCC (Foote 1997) using `librosa`, capturing key spectral characteristics of the speech signal. These MFCCs were extracted frame-by-frame and then zero padded or truncated to a fixed length of 300 frames to ensure uniform input dimensions.

4.2 Output Preparation

The models were trained on target emotions, which were derived from the IEMOCAP dataset by filtering out all other emotion categories and mapping "Excited" to "Happy" to maintain class balance and interpretability. This simplification enables a clearer comparison between model types while preserving the core affective distinctions relevant to SER.

4.3 Dataset Construction and Splitting

The resulting MFCCs and encoded labels were converted into NumPy arrays (with each MFCC’s matrix representation flattened when used in the RF model, but preserved as it is in the Bi-LSTM model to retain temporal information). To ensure balanced class representation, a stratified 80–20 train-validation split was applied based on the target emotions. The processed arrays were then wrapped into custom PyTorch Dataset classes, enabling efficient mini-batch loading during training.

4.4 Model Training

All three models were trained using a consistent 80/20 train-validation split with stratification to ensure fair class distribution. A random seed of 42 was used for reproducibility across experiments. The specific input representations were varied by model – flattened MFCCs for RF, padded MFCC

sequences for BiLSTM, and frozen embeddings for the Wav2Vec2 classification head. The training pipelines were standardised wherever applicable in terms of evaluation procedure and validation strategy, to ensure a consistent comparison.

RF: The RF model was trained using flattened MFCCs extracted from fixed-length audio segments. The dataset was filtered to include only the target emotions and split into stratified training and validation sets. The model was trained with 100 trees, using a fixed random seed for reproducibility. Performance was evaluated on the validation set with standard classification metrics, and a confusion matrix was generated to analyze class-wise results. This straightforward implementation serves as a baseline to assess the model’s capacity to discriminate emotions based on statistical acoustic features. As shown in Table 1, the model performed best on the Happy and Sad classes, while lower performance was observed for Angry and Neutral, indicating challenges in capturing less distinct emotional cues.

Class	Precision	Recall	F1-score	Support
Happy	0.46	0.71	0.56	527
Sad	0.56	0.57	0.57	250
Angry	0.62	0.28	0.38	254
Neutral	0.42	0.23	0.30	345
Accuracy			0.49	1376
Macro avg	0.52	0.45	0.45	1376
Weighted avg	0.50	0.49	0.46	1376

Table 1: Classification report for the RF model

Overall accuracy: 48.62%

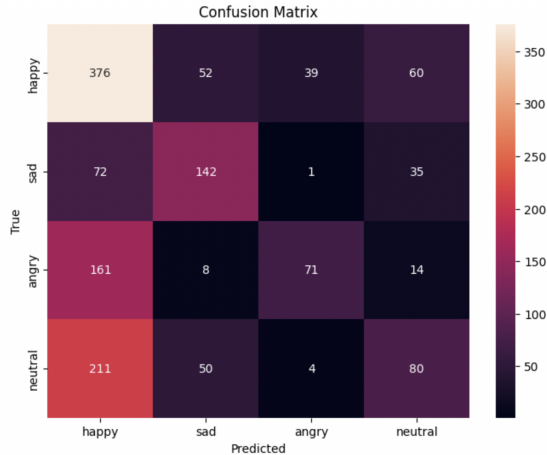


Figure 2: Confusion matrix for the RF model.

BiLSTM: The BiLSTM implementation pipeline (Yu & Kim 2020) begins with pre-processing the raw audio by extracting MFCCs with a fixed length of 300 frames and 13 coefficients per frame. As in the previous examples, emotions outside the four target emotions are filtered out, with Happy (including Excited) for consistency. The data is split into stratified training and validation sets. Each MFCC is padded or truncated to a fixed length to ensure batch compatibility, and a PyTorch dataset and dataloader pipeline is constructed to facilitate mini-batch training. The model is optimised using Adam with a learning rate of 1e-3 and trained using the categorical cross-entropy loss. A ReduceLROnPlateau scheduler adjusts the learning rate based on validation loss, and early stopping is employed with a patience of 10 epochs to avoid overfitting. The training loop, plotted in figure 3, includes tracking performance metrics and saving the best-performing model. As seen in Table 2, the model exhibited more balanced performance across classes compared to the RF model.

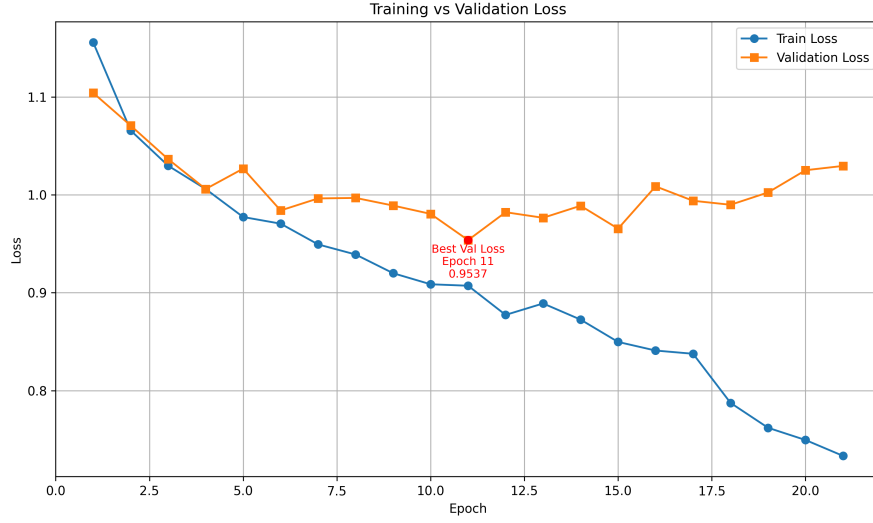


Figure 3: Training vs Validation Loss across epochs for the BiLSTM model. The lowest validation loss was observed at epoch 11, indicating the optimal model performance on unseen data.

Class	Precision	Recall	F1-score	Support
Happy	0.62	0.60	0.61	527
Sad	0.63	0.58	0.60	250
Angry	0.61	0.58	0.60	254
Neutral	0.48	0.54	0.51	345
Accuracy			0.58	1376
Macro avg	0.58	0.58	0.58	1376
Weighted avg	0.58	0.58	0.58	1376

Table 2: Classification report for the BiLSTM model

Overall accuracy: 58%

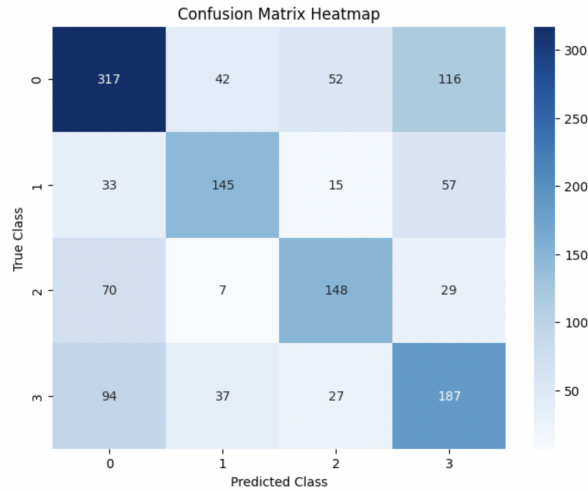


Figure 4: Confusion matrix for the BiLSTM model.

Wav2Vec2: The implementation pipeline for the Wav2Vec2-based model (Baevski et al. 2020) begins by extracting fixed-length embeddings from a pre-trained Wav2Vec2 model, which is kept frozen

throughout training. These embeddings are computed by mean-pooling the final hidden states for each utterance and cached to avoid repeated computation. The resulting feature vectors are then used to train a feed-forward classifier. The training process, plotted in figure 5, uses Adam Optimizer with a learning rate of 10^{-4} , and the objective is optimised using the negative log-likelihood loss. To improve generalization and manage learning dynamics, a step-based learning rate scheduler reduces the learning rate every 10 epochs. Early stopping is incorporated based on validation loss to prevent overfitting. As shown in Table 3, the model achieved competitive performance, particularly for the Angry and Sad classes, leveraging pre-trained acoustic representations.

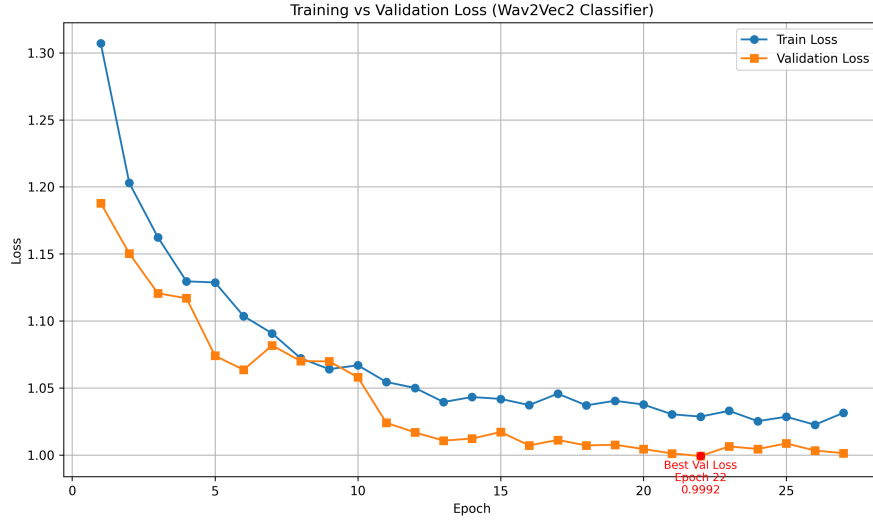


Figure 5: Training and validation loss across epochs for the Wav2Vec2-based classifier. The minimum validation loss is annotated.

Class	Precision	Recall	F1-score	Support
Happy	0.55	0.59	0.57	511
Sad	0.54	0.66	0.59	242
Angry	0.69	0.57	0.62	272
Neutral	0.47	0.42	0.44	351
Accuracy			0.55	1376
Macro avg	0.56	0.56	0.56	1376
Weighted avg	0.56	0.55	0.55	1376

Table 3: Classification report for the Wav2vec2 model

Overall accuracy: 55.31%

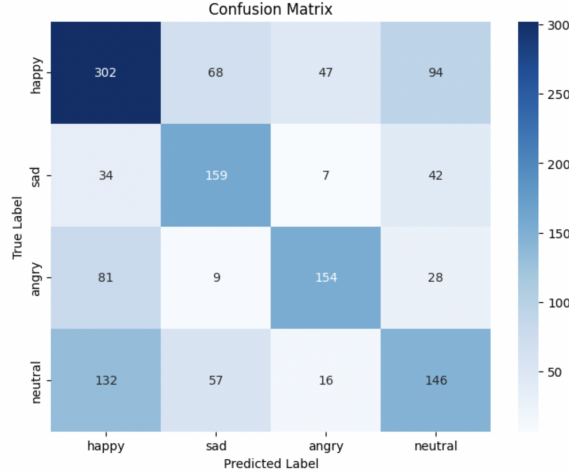


Figure 6: Confusion matrix for the Wav2Vec2 model.

4.5 Model comparison

The comparison of the three models reveals distinct strengths and weaknesses across different architectures. The RF model achieved the lowest overall accuracy at 48.62% and a UAR of 45%, with particularly low recall and F1-scores for the Angry and Neutral classes, indicating difficulty in capturing nuanced patterns from the flattened features, as expected. The BiLSTM model outperformed RF with an accuracy of 58% and a UAR of 58%, showing more balanced precision and recall across all classes, which highlights its ability to better leverage temporal dependencies in the data.

Meanwhile, the Wav2Vec2 model, which uses a frozen pre-trained feature extractor to obtain audio representations, over which a classification head is trained, achieved an accuracy of 55.31% and a UAR of 56%, slightly lower than the BiLSTM but still significantly better than the RF. Wav2Vec2 excelled in recognizing the Angry class with the highest F1-score of 62%, likely due to its powerful pre-trained audio representations, though it showed somewhat weaker performance on the Neutral class. Overall, the BiLSTM offered the most consistent and balanced performance across classes, while Wav2Vec demonstrated strong representation learning capabilities, and RF, as a classical baseline, fell short in handling the sequential and acoustic complexity of the task.

As illustrated in figures 2, 4, and 7, here we can see the different performances of all three models. We can clearly see how RF had very bad accuracy in general, specially for the two latter classes. The BiLSTM model shows much better results, specially handing the Neutral and Angry classes way better. Wav2Vec has somewhat similar results with the BiLSTM in general but it performs slightly better on the Sad and Angry categories and slightly worse on the Happy and Neutral.

4.5.1 Final Model Training on 8 Emotions

Following the initial experiments on the 4-class subset, the best-performing model architecture – namely the BiLSTM model – was retrained on the full 8-emotion classification task. This step was undertaken to capture the full spectrum of emotional granularity available in the IEMOCAP dataset. Training and evaluation protocols remained consistent with the prior setup to ensure comparability. As shown in Table 4, the results indicate a performance decline from the 4-class model due to increased class complexity and imbalance. The BiLSTM model achieved an accuracy of 42.19% and a UAR of 31% on the 8-class task.

Class	Precision	Recall	F1-score	Support
Neutral	0.33	0.44	0.38	345
Frustrated	0.42	0.50	0.46	584
Sad	0.47	0.59	0.53	250
Happy	0.17	0.09	0.12	131
Angry	0.57	0.40	0.47	254
Excited	0.47	0.35	0.40	395
Fear	0.00	0.00	0.00	22
Surprise	0.40	0.09	0.15	22
Accuracy			0.42	2003
Macro avg	0.36	0.31	0.31	2003
Weighted avg	0.42	0.42	0.41	2003

Table 4: Classification report for the best model trained on the full 8-emotion dataset

Overall accuracy: 42.19%

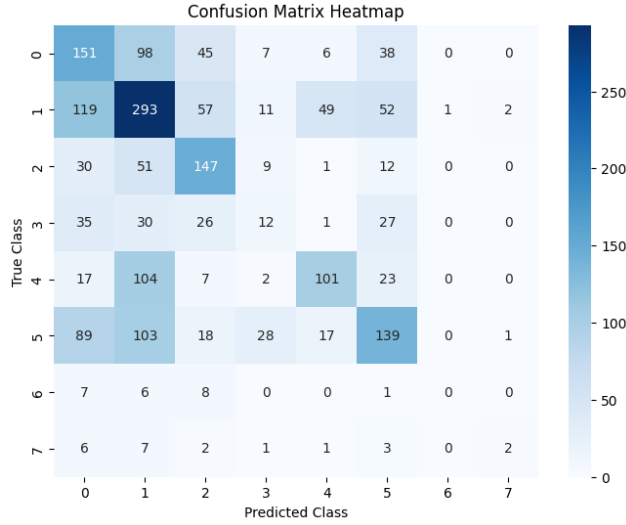


Figure 7: Confusion matrix for the BiLSTM model on 8-class subset.

5 Conclusions

The study conducted in our project helped us uncover several key insights, and identify the scope for future considerations.

5.1 Key Insights

Among all the models, the BiLSTM consistently performed the best. This suggests that modelling the temporal structure of speech is critical for effective emotion recognition. Recurrent neural networks are well-suited for this task because they can capture sequential dependencies across time, which are essential in expressing emotion. The study also indicates that MFCCs are effective representations of emotional content in speech. They capture pitch, tone, and spectral characteristics that often vary across different emotions.

5.2 Future Enhancements and Considerations

Looking ahead, several directions can be explored. One possibility is to represent emotions on a continuous scale rather than using discrete labels. This may better reflect the fluid and overlapping nature of real-world emotions. Another area is improving pre-processing strategies to enhance how

well models distinguish between similar emotional classes. Finally, incorporating multimodal data such as text could provide additional context. This can help models interpret emotional nuance more accurately.

5.3 Practical Implications

Overall, the study showcases the ability to use various learning models to automate SER. Such models, if fine-tuned for production use, can support applications in mental health monitoring, customer service, and human-computer interaction.

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